Worksheet — Work Done Simulation

Use with *Simulation - Work done.html* · work done by a force, area under F–s graphs and the work–energy principle (Block 9)

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/19-work-done/** in a browser (offline: *Simulation - Work done.html*). Choose the example at the top: **Angled pull · Lift vs ramp · Smart Cart · Braking**. Pick a preset in **Example** or set the values yourself: set a value with ▼ ▲ or type it in the box.

Press ▶ Start; ◂ / ▸ step through the motion. The middle column shows the free-body diagram and the work done by each force (or a Sankey diagram). On an *F–s* graph tick **Area**: the area is the work done.

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab (≈ 20 min) · **Part 3** homework (≈ 30 min). The Tasks tab has the same three parts. $W = Fs\cos\theta$ · $W_{net} = \Delta E_k$ · $E_k = \frac{1}{2}mv^2$. g = 9.81 N kg^{-1}. Answers to 3 s.f. with units.

Scan to open

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. 📐 A 5.0 N pull moves a box 1.0 m. The newton meter is at 0°, 30° or 60°. Which angle does the most work on the box?			
2. ⛰ Lift 1.0 kg up 0.30 m, or push it up a smooth 1.0 m ramp to the same height. Which needs more work? Which needs more force?			
3. You carry a bag horizontally at constant velocity. Is the work done by your upward force positive, negative or zero?			
4. 🚗 A car brakes with the same force from 20 m s^{-1} and from 40 m s^{-1}. How many times longer is the braking distance at 40 m s^{-1}?			
5. An elastic band is stretched twice as far, so its pull doubles too. How much more work does it do on a toy car?			

6. Calculate the work done by the 5.0 N pull at 60° over 1.0 m. Check: Angled pull → Example → Hook demo: 5.0 N at 60°.

Working

7. Calculate the force needed to push the 1.0 kg box up the smooth 1.0 m ramp (height 0.30 m) at constant speed, and the work done. Compare with lifting it straight up. Check: Lift vs ramp → Hook demo presets.

Working

8. (slide 8) A 1000 kg car at 20 m s^{-1} brakes with a force of 8000 N. Use the work–energy principle to calculate the braking distance.

Working

Part 2 — During the lab

Hook demo: pull at an angle. Simulation: Angled pull → Hook demo 0°, 30°, 60° (1.0 kg box, μ = 0.30, 5.0 N over 1.0 m). Press ▶️ Start, step to the end and copy the values from the middle column and the readouts. Write the newton-meter readings from the bench in the grey column.

angle θ	newton meter (bench) / N	F cos θ / N	F_N / N	F_f / N	W by F / J	W by friction / J	W_{net} / J	E_k / J
0°								
30°								
60°								

9. Why is the friction force smaller at 30° than at 0°? Why does the box still gain less kinetic energy at 60° than at 30°?

PASCO Smart Cart demo (lab 07). Simulation: Smart Cart → Class demo. Press ▶ Start; on the *F–s* graph tick Area and step to 0.60 m. The grey row has the class data (Capstone sample).

run	area under F–s, 0–0.60 m / J	E_k at 0.60 m / J	v at 0.60 m / m s^{-1}	difference / J	difference / %
simulation					
class data	0.475	0.444	1.33		

10. The kinetic energy is a little smaller than the work done by the pull. Explain why, and estimate the average friction force on the cart from the difference.

Working

11. Hook demo 3 (elastic band): use the two 'Elastic band' presets. Read the area under F–s for each and explain why doubling the stretch gives four times the work.

Working

Part 3 — Homework

12. (exit question) A 40 N force at 60° above the horizontal pulls a sled 5.0 m. Calculate the work done by the force.

Working

13. (exit question) 16 J of net work is done on a 2.0 kg trolley at rest. Calculate its final speed.

Working

14. (concept check) A box slides 2.0 m while a 5.0 N friction force acts on it. Calculate the work done by friction, with its sign.

Working

15. The slide 8 car (1000 kg) brakes from 40 m s^{-1} with the same 8000 N. Calculate the braking distance and compare it with question 8.

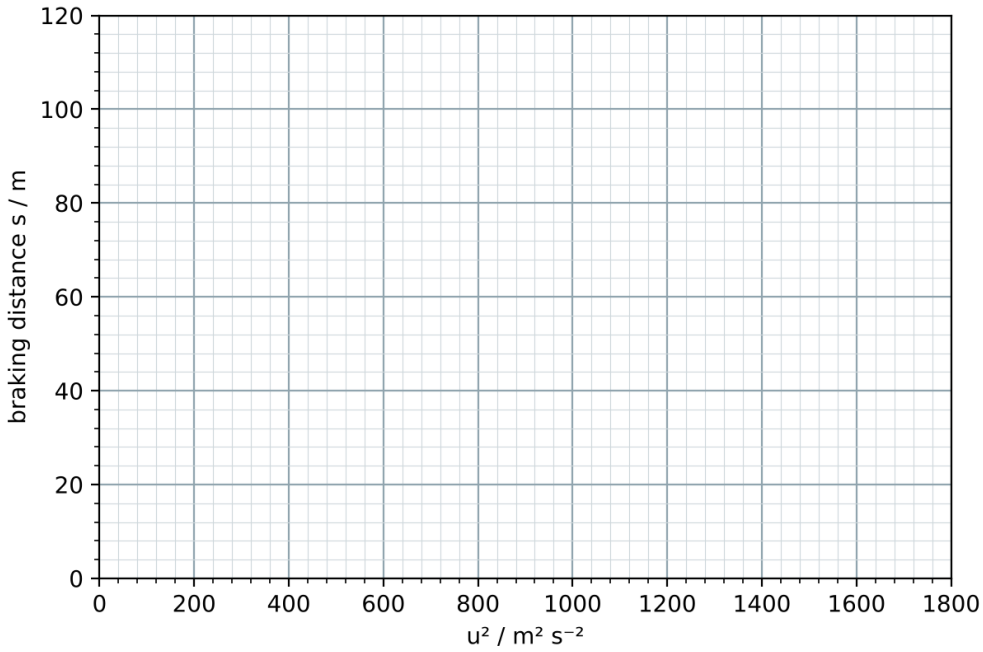
Working

16. A 1200 kg car must stop from 25 m s^{-1} within 50 m. Calculate the smallest braking force needed.

Working

17. Record the slide 8 car (1000 kg, 8000 N) at speeds from 10 to 40 m s^{-1} (Braking → set u, ● Record). Complete the table, plot s against u^2 and draw the best-fit line.

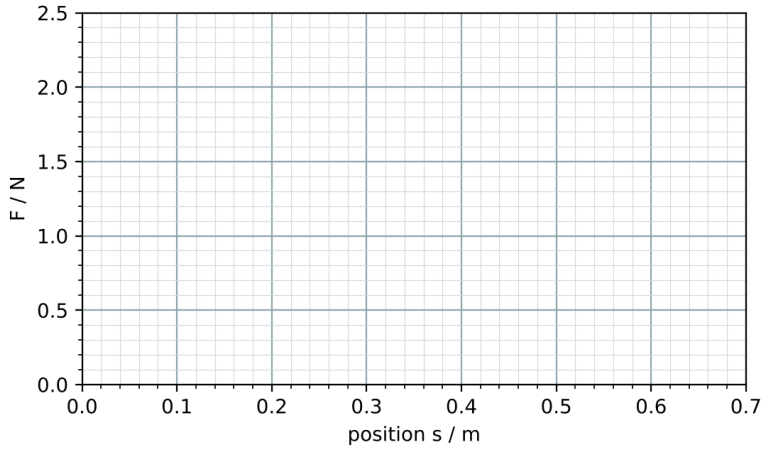
u / m s^{-1}	10	15	20	25	30	35	40
u^2 / m^2 s^{-2}							
s / m							



Gradient = _____ . Show that the gradient equals $m/(2F_b)$.

Working

18. Elastic band, stretch 2: the force rises steadily from 0.20 N to 2.0 N over 0.60 m. Sketch the F–s graph and calculate the work done. Check with the Area tool.



Working

19. (loading ramp) Explain why the normal force does no work on a box pushed up a ramp, and why a rough ramp needs more work than lifting the box straight up.